

Comparison

Sorting

Quick sort usually has a running time of $n \times \log(n)$, but is there an algorithm that can sort even faster? In general, this is not possible. Most sorting algorithms are comparison sorts, i.e. they sort a list just by comparing the elements to one another. A comparison sort algorithm cannot beat $n \times \log(n)$ (worst-case) running time, since $n \times \log(n)$ represents the minimum number of comparisons needed to know where to place each element. For more details, you

Problem

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Language Python 3

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```
1  #!/bin/python3
2
3  import math
4  import os
5  import random
6  import re
7  import sys
8
9  #
10 # Complete the 'countingSort' function below.
11 #
12 # The function is expected to return an INTEGER_ARRAY.
13 # The function accepts INTEGER_ARRAY arr as parameter.
14 #
15
16 def countingSort(arr):
17     # Write your code here
18     res=[0]*100;
19     for num in arr:
20         res[num]+=1
21     return res
22
23 if __name__ == '__main__':
24     fptr = open(os.environ['OUTPUT_PATH'], 'w')
25
26     n = int(input().strip())
27
28     arr = list(map(int, input().rstrip().split()))
29
30     result = countingSort(arr)
31
32     fptr.write(' '.join(map(str, result)))
33     fptr.write('\n')
34
35     fptr.close()
```

Line: 15 Col: 1

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✔ Test case 1 [🔒](#)

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